

Marios Papachristou
Assistant Professor of Information Systems
W. P. Carey School of Business
Arizona State University

LAST UPDATED	June 28, 2026
PERSONAL INFORMATION	E-mail mpapachr@asu.edu Phone +1-480-727-5771 Website papachristoumarios.github.io Office 400 E Lemon St, BAC 685, Tempe, AZ 85281 Nationality Greek, US Permanent Resident (Green Card Holder)
RESEARCH	My research develops models of resilience in AI-enabled socio-technical systems, focusing on how network structure, algorithmic agents, and information dynamics jointly determine systemic resilience of socio-technical systems.
METHODS	Network Science, AI Agents, Privacy, Computational Social Science
APPOINTMENTS	W.P. Carey School of Business, Arizona State University 2025 – Tenure-Track Assistant Professor, Department of Information Systems • <i>Affiliated Member</i>, Center for AI and Data Analytics (AIDA) • <i>Graduate Faculty Member</i>, Data Science, Analytics and Engineering • <i>Graduate Faculty Member</i>, Computer Science Department of Computer Science, Cornell University Summer 2026 Visiting Faculty
INDUSTRY EMPLOYMENT	Microsoft Research Summer 2023 Office of Applied Research, <i>with Chin-Chia Hsu and Longqi Yang</i> Twitter Summer 2022 User Modeling Research Group, <i>with Rishab Goel, Frank Portman, Matt Miller</i>
EDUCATION	Cornell University 2020 – 2025 Ph.D. in Computer Science, Minor: <i>Applied Mathematics</i> – <i>Doctoral Advisor</i>: Jon Kleinberg – <i>Thesis Title</i>: From Contagion to Stability: Insights into Network Dynamics, Resilience and Stability M.S. in Computer Science 2020 – 2022 National Technical University of Athens 2015 – 2020 Diploma in Electrical and Computer Engineering – <i>Advisor</i>: Dimitris Fotakis
HONORS & AWARDS	– ACM Doctoral Dissertation Award Nominee 2025 Nominated to the ACM by Cornell’s CS department as the best Ph.D. thesis in CS awarded in 2025. – ACM/AAAI SIGAI Doctoral Dissertation Award Nominee 2025 Nominated to the ACM SIGAI by Cornell’s CS department as the best Ph.D. thesis in AI awarded in 2025.

- **Sigma Xi** 2025
Elected member based on scientific contributions.
- **Cornell ORIE Young Researchers Workshop** 2024
- **Alexander A. Onassis Scholarship** 2023
Three-year fellowship for Greek Ph.D. students. Acceptance rate: 16%.
- **Bodossaki Foundation Fellowship** 2023
Runner-up.
- **LinkedIn Ph.D. Fellowship** 2022
One-year fully funded fellowship by LinkedIn. Acceptance rate 14%.
- **A.G. Leventis Scholarship** 2022
Award for graduate students of Greek origin. Acceptance rate 12.5%.
- **Gerondelis Scholarship** 2022
One-time award for graduate students of Greek descent.
- **Chateaubriand Fellowship** 2022
Fellowship granted by the French Embassy in the US for studies in France (declined for personal reasons). Acceptance rate 33%.
- **Cornell Fellowship** 2020
One-year fully funded fellowship by the CS department based on “exceptional preparation and promise”.
- **Thomaidion Award** 2019
Awarded by NTUA for publication during undergraduate studies.
- **ESEC/FSE 2019 ACM Student Research Competition Finalist** 2019
- **CanSat International Space Engineering Competition** 2019
4th out of 93 teams. Organized by the American Astronautical Society (AAS).
- **2nd prize “be finnovative 2.0” Start-up Accelerator by National Bank of Greece** 2018
Cashierless payments start-up.
- **1st prize at “Crowdhackathon Fintech #2” by the National Bank of Greece** 2017
Cashierless payments start-up.
- **IEEEExtreme 11.0 Programming Competition** 2017
Top 1% worldwide in IEEE’s algorithms competition.
- **Touramanoglu Scholarship** 2015
Highest University Entrance Exams score in the Imittos region.
- **“The Great Moment of Education” Scholarship by Eurobank SA** 2015
Highest University Entrance Exams score in the Imittos region.

GRANTS

- **ALCF Director’s Discretionary Allocation Award (DD Award)** 2026
Argonne Leadership Computing Facility. Project: Large-scale LLM-based Simulations for Algorithmic Auditing of Recommendation Systems

JOURNAL PAPERS

[Google Scholar Profile](#)

$\alpha\beta$ = alphabetical order

* = equal contribution

° = corresponding author

- [J1] **Network Formation and Dynamics among Multi-LLMs**
Marios Papachristou, Yuan Yuan
PNAS Nexus, 2025
[\[paper\]](#) [\[preprint\]](#) [\[code\]](#)
 - Taught at UC Berkeley, Stanford, UIUC and others
 - Featured by [TechXplore](#), and [SingularityHub](#)
- [J2] **Leveraging Large Language Models for Collective Decision-Making**
Marios Papachristou, Longqi Yang, Chin-Chia Hsu
Proceedings of the ACM on Human Computer Interaction (presented at ACM CSCW ’25), 2025
[\[paper\]](#)

- Invited talk at the LLMs as Research Tools Workshop at ACM CHI 2024
- Featured by the [ACM](#)

- [J3] **volesti: A C++ library for sampling and volume computation on convex bodies**
 ^{$\alpha\beta$} Apostolos Chalkis*, Vissarion Fisikopoulos*, *Marios Papachristou**, Elias Tsigaridas*
Journal of Open Source Software, 2025
[\[paper\]](#) [\[code\]](#)
- [J4] **Differentially Private Distributed Estimation and Learning**
Marios Papachristou, Amin Rahimian
IJSE Transactions (formerly IIE Transactions), 2025
[\[paper\]](#) [\[code\]](#) [\[talk\]](#)
 – Featured at the *July 2025 issue of the ISE Magazine* (10% of articles published in IJSE Transactions)
 – Featured by the *University of Pittsburgh Swanson School of Engineering*
- [J5] **Truncated Log-concave Sampling for Convex Bodies with Reflective Hamiltonian Monte Carlo**
 ^{$\alpha\beta$} Apostolos Chalkis, Vissarion Fisikopoulos, *Marios Papachristou*^o, Elias Tsigaridas
ACM Transactions on Mathematical Software (TOMS), 2023
[\[paper\]](#) [\[code\]](#)
 – Featured at [ODBMS.org](#)
- [J6] **Sublinear Domination and Core-periphery Networks**
Marios Papachristou
Scientific Reports, 2021
[\[paper\]](#) [\[code\]](#)
- [J7] **Commons-based peer production and digital fabrication: The case of a RepRap-based, Lego-built 3D printing-milling machine**
 Vasilis Kostakis*, and *Marios Papachristou**
Telematics and Informatics, 2014
[\[paper\]](#)
- TOP-TIER CS
 CONFERENCE
 PROCEEDINGS
- [C1] **Opinion Dynamics with Multiple Adversaries**
 Akhil Jalan*, *Marios Papachristou**^o
Forthcoming at WWW '26: ACM Web Conference (WWW), 2026
[\[paper\]](#)
 – Acceptance rate: 20%
- [C2] **Dynamic Interventions for Networked Contagions**
Marios Papachristou, Sid Banerjee, Jon Kleinberg
WWW '23: ACM Web Conference (WWW), 2023
[\[paper\]](#) [\[code\]](#)
 – Acceptance rate: 17%
- [C3] **Core-periphery Models for Hypergraphs**
Marios Papachristou, Jon Kleinberg
KDD '22: ACM Intl. Conf. on Knowledge Discovery and Data Mining (KDD), 2022
[\[paper\]](#) [\[code\]](#) [\[data\]](#)
 – Acceptance rate: 15%
 – Taught at tutorials at KDD and CIKM
- [C4] **Allocating Stimulus Checks in Times of Crisis**
Marios Papachristou, Jon Kleinberg
WWW '22: ACM Web Conference (WWW), 2022
[\[paper\]](#) [\[code\]](#)
 – Acceptance rate: 17%

- Featured by the [Federal Reserve Bank of Minneapolis](#)
- Taught at UC Berkeley

- [C5] **Software Clusterings with Vector Semantics and the Call Graph**
Marios Papachristou
 ESEC/FSE '19: ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE), 2019
[\[paper\]](#) [\[code\]](#) [\[data\]](#)
 – Acceptance rate: 21%
 – ESEC/FSE Student Research Competition Finalist Paper

REVISIONS & WORKING PAPERS

- [WP1] **The Power of Choice in Random Sampling**
Marios Papachristou, Jon Kleinberg
Major revision, Operations Research, 2025
[\[preprint\]](#)
 – Taught at the University of Pittsburgh
- [WP2] **Structural Measures of Resilience for Supply Chains**
Marios Papachristou, Amin Rahimian, Arash Azadegan
Major revision, Production and Operations Management, 2024
[\[preprint\]](#) [\[code\]](#)
- [WP3] **Echoes of Disagreement: Measuring Disparity in Social Consensus**
Marios Papachristou, Jon Kleinberg
Major revision, INFORMS Journal of Computing, 2024
[\[preprint\]](#) [\[code\]](#)
- [WP4] **Optimal Resource Allocation for Remediating Networked Contagions**
Marios Papachristou, Sid Banerjee, Jon Kleinberg
Under preparation for submission, 2025
[\[preprint\]](#)
- [WP5] **Differentially Private Distributed Inference**
Marios Papachristou, Amin Rahimian
Under preparation for submission, 2025
Preliminary version presented at the Theory and Practice of Differential Privacy Workshop (TPDP)
[\[preprint\]](#)

REFEREED CS WORKSHOPS

- [W1] **GLINKX: A Scalable Unified Framework for Homophilous and Heterophilous Graphs**
Marios Papachristou, Rishab Goel, Frank Portman, Matt Miller, Rong Jin
NeurIPS workshop on Graph Learning Frontiers (GLFrontiers), 2022
[\[paper\]](#) [\[poster\]](#)

TUTORIALS

- [T1] **Differential Privacy for Strategic Information Sharing and Learning: Foundations, Mechanisms, and Applications**
 Amin Rahimian, Juba Ziani, **Marios Papachristou**, Yuxin Liu
Conference on Web and Internet Economics (WINE), 2025

WORKS IN PROGRESS

- [WIP1] **Impact of Massive AI-generated Content on Recommender Systems**
 with Joonhwa Park, and Panos Adamopoulos
- [WIP2] **The Effects of Generative AI on Consumer Behavior**
 with Panos Adamopoulos, and Nikhil Garg

	[WIP3] Pivot, Commit, or Wait? Evaluating non-Committal Learning Strategies in Digital Ventures with Julian Lehmann, and Gerald Kane [WIP4] Ambient Language Models; Learning to Talk; Learning to Talk by Looking at Noise with Giannis Daras, Alkis Kalavasis, Costis Daskalakis, and Antonio Torralba [WIP5] Optimal Resolution of a Data Sharing Trilemma: Statistical Power, Sample Complexity and Privacy Budget with Yuxin Liu, Amin Rahimian, and Arash Azadegan [WIP6] The Price of Misreporting: Measuring the Effect of Adversarial Behavior in Online Platforms with Akhil Jalan
SOFTWARE	volesti github.com/GeomScale/volesti Open-source C++ statistical software for sampling and volume approximation from convex bodies.
OTHER EXPERIENCE	GeomScale Open-source Organization <i>May 2020 – May 2025</i> Sampling from truncated log-concave densities, and convex optimization, working on the volesti open-source package <i>with Dr. Apostolos Chalkis, Dr. Vissarion Fisikopoulos, and Dr. Elias Tsigaridas</i> Cashierless Payments Start-up <i>2017 – 2018</i> Part of National Bank of Greece’s start-up accelerator Athens University of Economics and Business <i>2018 – 2020</i> Department of Management Science and Technology, <i>with Prof. Diomidis Spinellis</i> Google Summer of Code 2020 (GeomScale) <i>June 2020 – August 2020</i> Sampling from high-dimensional truncated log-concave densities, <i>with Dr. Apostolos Chalkis, Dr. Vissarion Fisikopoulos, and Dr. Elias Tsigaridas</i> [code] [talk] Google Summer of Code 2018 (GFOSS-OTA) <i>April 2018 – September 2018</i> Automated codification of Greek Legislation, <i>with Prof. Diomidis Spinellis</i> [code] [data] [talk] Researcher – P2P Lab <i>2013 – 2014</i>
TEACHING	Arizona State University, Instructor <i>Spring 2026</i> CIS 509: Analytics for Unstructured Data – Evaluation (out of 7.0, MSBA/MSBA/MSAIB): 6.8/6.6/6.4 (Spring 2026) Cornell University, Instructor <i>Spring 2023, Fall 2023</i> CS 6850: The Structure of Information Networks (Ph.D.-level) – Evaluation (out of 5.0): 5.0 (Spring 2023), 4.8 (Fall 2023)
MENTEES	– Kyuyeon (Jenny) Rhee (Ph.D. student at ASU) – Pin-Jun Liu (Ph.D. student at ASU) – Joonhwa Park (Ph.D. student at Emory) – Yuxin Liu (Ph.D. student at the University of Pittsburgh)

- Jasmine Li (Undergraduate Cornell CS)
- Ioannis Iakovidis (Google Summer of Code 2022; now Ph.D. student at Johns Hopkins)
- Zhang Zhuyan (Google Summer of Code 2022; now Technology analyst at Bank of America)

MENTORSHIP EXPERIENCE

- Student-applicant Support Program at Cornell CIS 2022
- Google Summer of Code Mentor 2019 – 2022
- Volunteer lessons for the National Olympiad of Informatics (PDP) 2019

TALKS & LECTURES

[†]= scheduled

*= presented by co-author

- ***Resilience in Networks: Frameworks for Resource Allocation and Simulation***
 - Archimedes, Athena RC April 2025
 - TOC4Fairness Seminar, Simons Institute March 2025
 - Cornell University, Theory Seminar December 2024
 - Arizona State University, W.P. Carey School of Business November 2024
 - Purdue University, Daniels School of Business November 2024
 - MIT, Research Group Meeting June 2024
- ***Network Formation and Dynamics among Multi-LLMs***
 - Arizona State University, School of Mathematical and Natural Sciences, AI Seminar November 2025
 - INFORMS Annual Meeting October 2025
 - Symposium on Statistical Challenges on Electronic Commerce Research (SCECR) June 2025
 - Arizona State University, Machine Learning Day April 2025
 - NSF Conference on Network Science and Economics April 2025
 - INFORMS Annual Meeting October 2024
 - IBM Research, Exploratory Math Sciences Council Seminar April 2024
 - UIUC, Guest Lecture at the Network Analysis Course April 2024
 - Cornell University, AIPP Group March 2024
 - Learning on Graphs (LoG) NYC Meetup February 2024
- ***Information Aggregation and Distributed Learning in Privacy-critical Environments***
 - CMU, Tepper School of Business OM Seminar* September 2024
 - TOC4Fairness Online Seminar Series* March 2024
 - Information: Theory and Applications (ITA) Workshop* February 2024
 - Rutgers University Business School, Seminar* November 2023
- ***Leveraging Large Language Models for Collective Decision-Making***
 - LLMs as Research Tools: Applications and Evaluations in HCI Data Work Workshop, ACM Conference on Human-Computer Interaction* May 2024
 - Microsoft Research, Invited Applied Research Talks August 2023
- ***Structural Measures of Resilience for Supply Chains***
 - NSF Conference on Network Science and Economics April 2024
 - Columbia University, Student Theory Seminar November 2023
 - Cornell University, Theory Tea November 2023
 - INFORMS Annual Meeting* October 2023
 - Cornell University, LinkedIn Campus Visit May 2023
- ***Resource Allocation in a Financial Contagion Environment***
 - University of Chicago, Theory Seminar October 2023
 - Indiana University, Center for Complex Networks and Systems Research September 2023
 - Cornell University, CS 6850 Guest Lecture April 2023
 - Cornell University, Theory Seminar November 2022
- ***Dynamic Interventions for Networked Contagions***
 - INFORMS Annual Meeting October 2023

- ACM Web Conference April 2023
- ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization October 2022
- ***Core-periphery Models for Hypergraphs***
 - ACM Conference on Knowledge Discovery and Data Mining August 2022
- ***GLINKX: A Scalable Unified Framework for Homophilous and Heterophilous Graphs***
 - NeurIPS Workshop on Graph Learning Frontiers December 2022
 - Twitter, Machine Learning Seminar August 2022
- ***Allocating Stimulus Checks in Times of Crisis***
 - ACM Web Conference April 2022
 - ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization October 2021
- ***Sampling from Truncated High Dimensional Logconcave Densities***
 - PyData Global December 2020
- ***Software Clusterings with Vector Semantics and the Call Graph***
 - ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering August 2019

DEPARTMENTAL
AND
PROFESSIONAL
SERVICE

Department and School Outreach

- AI and Data Analytics Center (AIDA) Brown Bag Seminar Series Co-organizer [\[link\]](#)
- Department of Information Systems Brown Bag Seminar Co-organizer [\[link\]](#)
- W. P. Carey School of Business, PhD Lunch and Learn, November 2025, Keynote Speaker [\[link\]](#)

Program Committee

- ACM Conference on Fairness Accountability and Transparency (FAccT)
- ACM Conference on Knowledge Discovery and Data Mining (KDD)
- ACM Conference on Economics and Computation (EC)
- IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS)

Ad-hoc Reviewer

- MIS Quarterly
- Management Science
- Mathematics of Operation Research
- ACM Transactions on Management Information Systems
- Journal of Artificial Intelligence Research
- Naval Research Logistics
- International Conference on Computational Social Science (IC2S2)
- ACM Web Conference (WWW)
- Innovations in Theoretical Computer Science (ITCS)
- ACM Conference on Human Factors in Computing (CHI)
- European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD)
- Machine Learning
- NeurIPS 2022 Workshop on Graph Learning Frontiers (GLFrontiers)
- NeurIPS 2021 Workshop on Human and Machine Decisions
- Journal of Open Source Software

Session Chair

- INFORMS Annual Meeting 2025
- NSF Conference on Network Science and Economics 2024
- ACM Web Conference (WWW) 2022

REFERENCES

Available upon request.